Svaraa Fold-over Edition — Hardware Specification

Oct 9, 2026 · Svaraa Ventures Inc.

Summary

Svaraa Fold-over Edition (SL1-F) is the Svaraa Lite pendant split into two discs joined by a flexible TPU bridge: folded shut it hangs on a cord, folded over a collar or pocket edge it clips on with enclosed magnets. Electronics, firmware, app and cloud are Svaraa Lite SL1 v1.0; the edition adds a fold-state Hall sensor and a physical microphone privacy switch, and keeps USB-C charging.



Item	Value
Issued by	Svaraa Ventures Inc.
Author	Vishwajeet J K
Product code	SL1-F (single SKU: pendant, braided cord, USB-C cable)

Item	Value
Base platform	Svaraa Lite SL1 v1.0 (23 Sep 2026): same PCB function, firmware, app and cloud
Look	Aubergine matte polymer face with gold lotus, flush matching button, dusty-rose TPU body, aubergine rear pad, cream braided cord
Envelope	Main body Ø42.0 × 9.6 mm, rear half Ø42.0 × 3.0 mm; ≈ 12.6 mm folded shut; ≈ 30 g; IP67 (estimates)
Battery life	≈ 4–5 days typical, ⩾ 2 days heavy use (estimates, as SL1)
Charging	Waterproof USB-C at 6 o′clock, any standard cable and charger, ≈ 2 h to full
Ways to wear	Closed on the cord, or clipped over any fabric edge up to 4 mm thick
BOM target	≈ $50 at 10k units including cord and cable (estimate)
Programme	EVT 8–10 weeks from kickoff · MP ≈ 7–8 months

Like SL1, the Fold-over Edition uses only standard, commercially available components and methods and does not include the proprietary platform features of the full Svaraa line. Numbers marked "estimate" are engineering estimates for partner review. Anything not restated here follows the SL1 v1.0 specification.

1 Edition at a glance

The Fold-over Edition keeps SL1′s electronics, USB-C charging, battery and cost, and trades the steel bezel and mala for a two-part polymer body that also clips onto clothing. The price is thickness: ≈ 12.6 mm folded shut against SL1′s 9.8 mm.

Area	Svaraa Lite (SL1)	Fold-over Edition (SL1-F)
Form	Single disc on a bail and rudraksha mala	Main body and rear half joined by a TPU bridge; folds shut as a pendant or over fabric as a clip
Enclosure	316L PVD rose-gold bezel, 0.8 mm printed glass face	PC + glass-fibre core with dusty-rose TPU overmould; matte aubergine polymer face

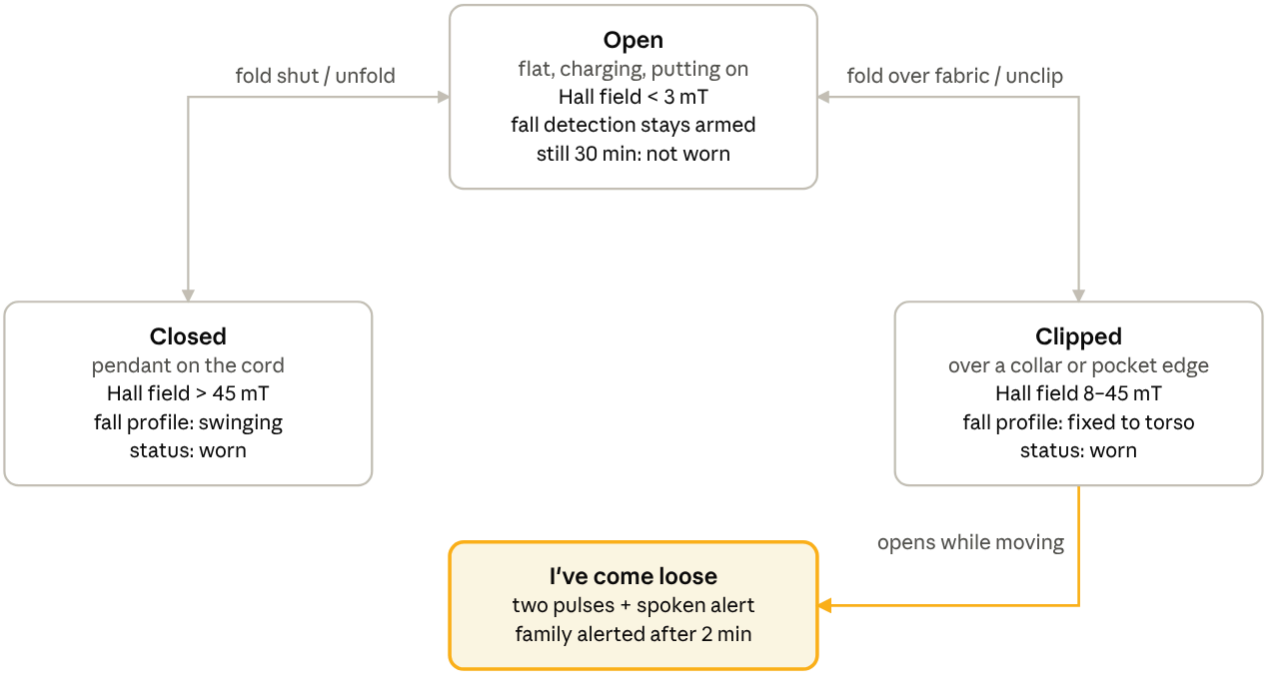
Area	Svaraa Lite (SL1)	Fold-over Edition (SL1-F)
Button	Ø6 mm gold PVD cap, 0.3 mm proud	Flush aubergine cap matching the face, printed gold ring and 0.2 mm raised rim
Mic privacy	Triple press cuts mic power	Physical slider at 5 o′clock; hardware cuts mic power and firmware cannot override it
New sensing	—	Fold-state Hall sensor: closed, clipped or open
Charging	Waterproof USB-C at 6 o′clock	Same
Attachment	Integral bail	Cord through a loop moulded into the bridge, or clipped over a fabric edge up to 4 mm
Size	Ø42 × 9.8 mm	Ø42 mm; 9.6 mm main body + 3.0 mm rear half = ≈ 12.6 mm folded
Weight	≈ 30 g	≈ 30 g
Battery	330 mAh, ≈ 4–5 days typical	Same
BOM (10k, estimate)	≈ $50 pendant + cable	≈ $50 pendant + cord + cable
Sealing	IP67 with the port open	IP67 main body; the rear half has no electronics

The rear half and bridge carry no wires: the Hall sensor sits on the main PCB and reads the rear-half magnets through the back cover, so nothing electrical crosses the part that flexes.

2 How the fold works

A single Hall sensor on the main PCB reads the rear-half magnets through the back cover, and the field strength alone tells the pendant whether it is closed, clipped over fabric or open; that one reading sets the fall profile, wear status and the loose-clip alert with no setup by the wearer.

One Hall sensor tells the pendant if it is closed, clipped or open



Fold states · 3 Hall bands, 1 loose-clip alert

Band thresholds are estimates to be set at EVT; hysteresis between bands stops a clip on thin fabric flickering between clipped and closed.

3 The seven R's on the Fold-over Edition

All seven R's work as in SL1; Respond, Rescue and Repository change because the fold state is now known automatically and the microphones have a physical switch.

R	What changes on SL1-F
01 Recite	Unchanged audio path. Speaker holes move from the bezel into the face at 12 o'clock.
02 Respond	Single press still talks to Svaraa. With the privacy slider off, a press answers "Microphones are switched off" instead of listening.
03 Remind	Unchanged: schedule and real-time clock on the pendant, so reminders fire offline.

R	What changes on SL1-F
04 Rescue	Fall profile follows the fold state with no setup: closed on the cord uses SL1′s swinging-pendant thresholds, clipped uses a fixed-to-torso profile. A clip that opens while the wearer is moving triggers "I′ve come loose" (section 2). With the mics off, the countdown is cancelled by a press only.
05 Recall	Unchanged: cloud transcription and search.
06 Repository	Double press starts and stops recording (the render′s "one-press recording"), with SL1′s start, stop and 10-minute chimes. Sliding the mics off ends a recording at once.
07 Riddle	Unchanged: all integrations live in the app and cloud.

4 Electrical architecture & components

The SL1 block diagram carries over unchanged, including the waterproof USB-C receptacle with USB 2.0 data; SL1-F adds two Hall sensors on the main PCB and puts the microphone load switch under hardware control of the privacy slider.

Function	Reference part	Requirement / notes	Alternates
Fold-state sensor	TI TMAG5273 3D linear Hall sensor (±80 mT range variant), SOT-23-6, I2C + interrupt	On the PCB underside, under the rear-half sense magnet; reads it through battery and back cover; bands: closed > 45 mT, clipped 8–45 mT, open < 3 mT (estimates, set at EVT); wake-on-threshold at 5 Hz, ⩽ 10 µA average	Two TI DRV5032 Hall switches with different thresholds
Privacy switch sensor	TI DRV5032 unipolar Hall switch, 20 Hz variant	Under the slider at 5 o′clock; output gates the mic load switch in hardware; ⩽ 100 ms from slide to mic power off	IP67-rated miniature slide switch breaking mic VDD directly

Function	Reference part	Requirement / notes	Alternates
Mic load switch	TI TPS22916-class load switch	Enable = slider ON AND SoC request; firmware reads the slider on a GPIO but cannot override it	Discrete P-FET
Slider magnet	Ø2 × 1 mm NdFeB N48H, captive in the slider	Present over the sensor only in the ON position	—
Steel targets	2 × Ø12 × 0.4 mm 430 stainless, in the main back cover	Give two rear-half magnets something to pull on; the third (sense) magnet faces the Hall sensor with no target in the way	Magnet pairs (stronger hold, more stray field)
Housing	Polymer core; no RF window needed	Chip antenna retuned at EVT for the TPU overmould and the rear half folded behind it	—

All other parts (nRF5340, TDK T5838 mics, LSM6DSV16X IMU, BMP581 barometer, 1 Gb QSPI NOR, MAX98357A, 15 × 11 × 3 mm speaker, LRA + DRV2605L, nPM1300, waterproof USB-C with TPS2595-class OVP, Ø35 × 2.7 mm 330 mAh cell) follow the SL1 component table. Speaker, LRA and attachment magnets shift the Hall baseline, so each unit's open and closed readings are calibrated at end-of-line test.

5 Mechanical design: body, rear half & bridge

The bridge sits at 12 o'clock of the main body, so in both wear modes it is the top edge: it carries the cord when closed and hooks over the fabric edge when clipped, which puts the pendant's weight on the bridge and leaves the magnets only to stop it peeling open. USB-C stays at 6 o'clock, the bottom edge, as in SL1.

Main body layer (front to back)	Thickness
TPU overmould lip over the face edge	0.3 mm
Matte PC face, in-mould aubergine + gold lotus, speaker holes at 12 o'clock	1.0 mm

Main body layer (front to back)	Thickness
Front cavity: button dome + membrane, acoustic ducts, speaker/LRA protrusion	1.2 mm
Main PCB zone: Ø36 mm 6-layer HDI, mid-mount USB-C, speaker and LRA in cutouts	3.4 mm
Battery: Ø35 × 2.7 mm round Li-Po, 330 mAh	2.7 mm
Back cover with two steel targets, O-ring	1.0 mm
Main body total	9.6 mm (10.0 design-to)

Rear half layer (main-body side to skin side)	Thickness
Base shell, PC + glass fibre inside the TPU overmould	0.8 mm
Three Ø12 × 1.0 mm N48H magnets, each backed by a 0.4 mm steel shunt disc on the skin side	1.4 mm
Rear pad: aubergine textured TPU	0.8 mm
Rear half total	3.0 mm

Part	Specification
Bridge	Same 2-shot TPU overmould as both halves (Shore 85A, dusty rose), 26 mm free length, 14 mm wide, 1.6 mm thick, embedded woven polyester tape; folded shut, the spare length forms an arch above the pendant
Cord slot	6 × 4 mm slot through the bridge at its midpoint; the cord can stay threaded in clip mode as a tether round the neck
Main housing	PC + 10 % glass-fibre core; mic ports at 10 and 2 o′clock behind ePTFE membranes; USB-C in a flat at 6 o′clock; slider track at 5 o′clock
Face and button	1.0 mm matte PC with in-mould decoration; Ø6 mm flush aubergine cap with a printed gold ring and 0.2 mm raised rim; same dome + membrane as SL1 (1.8–2.5 N, 200k cycles)

Part	Specification
Rear pad	Aubergine textured TPU, the only part against the skin in both wear modes; magnets fully overmoulded, no openings
Cord	Ø3.5 mm braided cream polyester, 60–80 cm adjustable, breakaway clasp releasing at 15–25 N
Sealing	IP67 main body with the sealed USB-C open; rear half and bridge have no electronics
Mass	≈ 30 g: main body ≈ 22 g, rear half ≈ 7 g, bridge ≈ 1 g (estimates)

Fold and attachment requirements (to be measured at EVT):

- Closed with no fabric: ⩾ 10 N to pull the halves apart, so a swinging pendant never opens by itself.
- Clipped over 2 mm of fabric: ⩾ 5 N peel; over 4 mm: ⩾ 2.5 N; fabric up to 4 mm fits the bridge.
- Bridge: ⩾ 20,000 fold cycles from −10 to +45 °C with no cracking or whitening; cord slot tear ⩾ 60 N, above the 15–25 N cord breakaway; bridge-to-body peel ⩾ 80 N.
- No magnet comes loose after 1.5 m drops onto concrete and a 50 N push-out test.

The concept render shows the bridge at the side of the main body in its open view; this spec fixes it at 12 o′clock so the same edge serves as cord loop and clip hook.

6 Physical microphone privacy switch

A slider at 5 o′clock, beside the USB-C port, removes power from both microphones in hardware; firmware can read its position but cannot turn the mics back on. It replaces SL1′s triple-press mute as the wearer′s main privacy control.

The slider is an aubergine polymer tab in a track moulded into the TPU rim, with 3 mm of travel and a detent you can feel at each end. It carries a Ø2 × 1 mm magnet: in the ON position the magnet sits over a Hall switch on the PCB, whose output enables the mic load switch. Slid to OFF, the magnet moves away and mic power drops within 100 ms. No hole passes through the housing, so the main body stays IP67 without a moving seal.

The design is fail-private: if the slider magnet is missing, cracked or out of place, the microphones stay off.

Action	Feel / hear / see
Slide to OFF	Detent click + "Microphones are off"; a printed crossed-microphone icon is uncovered
Slide to ON	Detent click + "Microphones are on"
Press to talk with mics off	"Microphones are switched off. Slide the switch to talk."
Status request	Includes "microphones off" when they are
App	Family can see the mic state but cannot change it remotely

With the mics off, the button, SOS, fall detection and countdown, reminders and audio playback all keep working; talking, recording, the optional wake word and the spoken "I'm fine" do not.

Requirements: slide force 2–4 N, operable with a fingertip and no fingernail; ≥ 10,000 cycles; state unaffected by the rear-half magnets in every fold state and by a 5 mT external field from any direction (EVT test). Alternate if the magnetic slider fails EVT: an IP67-rated miniature slide switch breaking mic VDD directly.

7 Power, USB-C charging & battery life

Battery life matches SL1, ≈ 5 days typical when new: the two Hall sensors add about 0.3 mAh/day (fold sensor ≤ 10 µA average, privacy switch ≈ 1 µA) against a ≈ 60 mAh/day budget.

Use profile	mAh/day	New (297 mAh usable)	End of life (80 %)
Light (few questions, no recording)	≈ 35	≈ 8 days	≈ 6.5 days
Typical (SL1 daily profile + Hall sensors)	≈ 60	≈ 5 days	≈ 4 days
Heavy (wake word 16 h + continuous recording 16 h)	≈ 155	≈ 1.9 days	≈ 1.5 days

All figures are engineering estimates from datasheet currents; EVT replaces them with measured values per mode.

Charging is SL1's: any standard USB-C cable and charger, 200 mA CC/CV to full in ≈ 2 h, JEITA derated above 40 °C and suspended above 45 °C, ship mode ⩽ 2 µA woken by plugging in. The port is reachable with the pendant open or closed, and the cord can stay on while charging. The SL1 charging stand needs a deeper cradle for the 12.6 mm folded thickness. Low-battery behaviour is unchanged.

8 User interface without lights

With mute moved to the physical slider, the button map frees a gesture for one-press recording; everything else is said aloud or felt, as in SL1.

Press	Action
Single press	Talk to Svaraa; also acknowledges a reminder
Hold 3 s	Call for help (SOS); vibration pulses count down, press again to cancel
Double press	Start or stop a recording, with rising and falling chimes
Triple press	Status: "Svaraa is ready, battery 70 percent, clipped on, microphones on, connected to your phone"
Hold 10 s while plugged in	Restart (service only)

Only single press and hold need to be taught; double and triple press can be disabled or swapped in the app. Single press stays as Talk, so the render's "one-press recording" is a double press.

Event	Feel / hear
Folded over fabric	One short tick (confirms the clip has seated)
Clip opens while moving	Two quick pulses + "I've come loose"
Open and still for 30 min	Nothing on the pendant; the app shows "not being worn"
Mics switched off / on	Detent click + "Microphones are off" / "on"

The flush, face-coloured button is harder to find than SL1's gold cap, so it gets a printed gold ring, a 0.2 mm raised rim and a smoother texture than the matte face; the DVT home trial checks that wearers find it without looking at least 19 times in 20. All SL1 vibration

and sound patterns (reminders, SOS countdown, help sent, out of range, charging) are unchanged.

9 Safety: falls, magnets & cord

The SL1 fall and SOS method stays, now tuned automatically by fold state; the main new risk is that the magnet half always faces the body, so its field toward the skin is capped.

9.1 Fall and SOS by fold state

Fold state	Fall profile	Notes
Closed on the cord	SL1 method unchanged: free-fall ≥ 150 ms then impact ≥ 2.5 g; rotation ≥ 200 °/s and tilt ≥ 60°; 20 s stillness	SL1 thresholds were set for a swinging pendant
Clipped over fabric	Same method; tilt measures the torso directly, so thresholds may be tightened from DVT data	Fixed mounting gives cleaner signals and fewer false alarms
Open	Stays armed; a fall candidate is handled normally	Open and still for 30 min marks "not being worn" in the app
Charging (USB-C connected)	Paused	Hold-3-s SOS still works

Hold-3-s SOS, the 30 s "Did you fall?" countdown and the out-of-range loud alarm are the same in every state. With the privacy slider off, the countdown can only be cancelled by a press, and the app tells family that voice confirmation was unavailable.

9.2 Magnets and cardiac implants

Pacemakers and ICDs are tested to tolerate static fields only up to 1 mT (10 gauss) for temporary malfunction (ISO 14117:2019, clause 4.6). In this design the rear half faces the body in both wear modes, and by our estimate three bare Ø12 × 1.0 mm N48H magnets would still exceed 1 mT a few centimetres out, so each magnet is backed by a steel shunt disc on the skin side.

- Requirement: ≤ 1 mT at 20 mm from the rear pad surface, closed and clipped, measured with a calibrated gaussmeter at EVT and DVT.
- Magnet polarity mixed (two one way, one the other) to cut the far field further.

- The app asks at setup about a pacemaker or ICD; if yes, it advises against clipping on the left chest or a left breast pocket, suggests the cord or a waistband, and a cardiac-device consultant reviews the field data before launch.
- Leaflet and box carry an implant warning, with the separation distance set from the measured field.

9.3 Cord, pinch and small parts

- The cord breakaway (15–25 N) releases well before the bridge slot can tear (≥ 60 N), so the pendant cannot become a strangulation hazard on a bed rail, walker or door handle.
- The halves snap shut with up to ≈ 10 N; all edges that meet are rounded to ≥ 1 mm radius so the snap cannot pinch thin skin hard.
- Magnets are fully overmoulded and the rear half is far larger than the small-parts cylinder; drop and push-out tests cover every magnet.

Location (from the phone's GPS), lost-pendant finding (BLE, Find My or Find Hub) and integrations are unchanged from SL1; the "I've come loose" alert adds the last known location to the app.

10 Firmware & software integration

Firmware, app and cloud are the SL1 stack (Zephyr on nRF5340, MCUboot, standard BLE services plus the encrypted Svaraa Lite control service, USB 2.0 data for service); SL1-F adds fold-state logic and reports the privacy slider.

Item	Change from SL1
Fold-state driver	TMAG5273 in wake-on-threshold mode; debounced bands for closed, clipped and open with hysteresis; per-unit calibration stored at end-of-line
Fall profiles	Chosen automatically from fold state; thresholds held in settings and updatable over the air
Loose-clip logic	Clipped → open while the IMU shows walking: pulses + spoken alert at once; app alerts family with last location if not re-clipped within 2 min
Events (notify)	New: fold state changed, came loose, privacy slider on/off; existing events unchanged
Control commands	SL1 "mute" stays as a soft mute; it can never switch mics on while the slider is off

Item	Change from SL1
Button map	Double press = record, triple press = status; both remappable in the app
App	Shows fold state, wear time and mic state; setup adds the cardiac-implant question (section 9.2) and a short clip-on demo

The SL1 reliability rules are unchanged: reminders, fall detection and the SOS countdown never depend on the phone, events are stored until the app confirms receipt, OTA runs only while charging with battery ≥ 50 %, and family is warned if the pendant has not connected for 2 h during the day or has not been charged for 3 days.

11 Certification, testing, cost, timeline & risks

The radio and charging hardware are SL1's, so the certification plan is unchanged; the pendant with cord and cable lands at ≈ $50 at 10k units (estimate), the same as SL1, because the polymer body offsets the new sensors and rear half.

Certifications: SL1 set unchanged (FCC Part 15C + 15B, ISED RSS-247, WPC ETA, BIS CRS, CE RED incl. EN 18031, UKCA, Bluetooth SIG, UN 38.3, IEC 62133-2, RoHS, REACH, WEEE). The rear pad and bridge, which touch skin, are tested to ISO 10993-5 and -10.

Production tests (100 %): the SL1 list plus Hall calibration open and closed in a fixture, mic power confirmed off with the slider at OFF, slider detents, and a visual check of the bridge. Fold hold force is sampled per lot.

Qualification: the SL1 list (IP67, 1.5 m drop, −10 to +45 °C, 85 % RH 40 °C 96 h, 200k button presses, 10k USB-C insertions, 500 battery cycles, ESD ±8 kV) plus 20,000 fold cycles at −10 and +45 °C; bridge slot tear and bridge peel; cord breakaway over 1,000 cycles; 10,000 slider cycles; magnet push-out; UV ageing of the TPU; artificial sweat on the rear pad and slider track; stain tests on the dusty-rose TPU with turmeric, kumkum and sandalwood paste; static-field mapping per section 9.2.

BOM item (10k units, estimate)	USD
nRF5340 + crystals + antenna + RF passives	9.0
2 mics, IMU (accel + gyro), barometer	6.0
Amp, speaker, LRA + driver	5.0
NOR flash, PMIC, charge protection	5.5

BOM item (10k units, estimate)	USD
Battery 330 mAh round	3.0
PCB 6-layer HDI + assembly + test	8.5
Fold sensor, privacy Hall switch, mic load switch	1.0
Main body: core, face, button, back cover, steel targets, membranes, slider	3.7
2-shot TPU overmould (body, bridge, rear shell)	2.0
Rear half: pad, 3 magnets, shunt discs	1.8
Waterproof USB-C receptacle, cable, packaging	2.5
Braided cord with breakaway clasp	1.5
Pendant + cord + cable	≈ 50 (45–60)
NRE: tooling (incl. 2-shot bridge tool) + design + certification	≈ 130–190k

Timeline, from kickoff:

1. EVT, 8–10 weeks: 50 units, all functions, power, Hall bands and magnetic field measured; bridge soft tool.
2. DVT, 8 weeks: 200 units; final mechanics and reliability including the 20,000-cycle fold test; 20-elder home trial in both wear modes across common garments (shirt, kurta, sari blouse, dupatta, cardigan); pre-scan.
3. PVT, 6 weeks: 1k units; certifications complete; yield ⩾ 95 %.
4. MP: ≈ 7–8 months from kickoff.

Risk	Mitigation
Bridge cracks or whitens from daily folding	TPU grade and embedded tape; 20,000-cycle test at temperature extremes; generous bend radius from the 26 mm free length
Magnetic field near a pacemaker or ICD	Steel shunt discs, field limit and measurement (9.2); setup question; consultant review; labelling
12.6 mm folded feels bulky	Thinner rear half (2.5 mm with Ø12 × 0.8 mm magnets) if EVT hold force allows; elder feedback at DVT

Risk	Mitigation
Clip slides off silk or slips on thick wool	Bridge carries the weight over the fabric edge; 4 mm fabric limit; "I've come loose" alert; cord as tether
Flush, face-coloured button hard to find	Printed gold ring, raised rim, texture; findability checked in the home trial
Privacy slider misread by stray magnets	Fail-private design; 5 mT immunity test; sealed slide switch as fallback
Hall bands drift with temperature or between units	Per-unit calibration, hysteresis, the sensor's temperature compensation
Third-party patents on fold-over magnetic clips and fall detection	Freedom-to-operate review before launch, alongside the SL1 review